Hospital Recommendations Based on Statewide Planning and Research Cooperative System Data for Medicare Patients

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Abstract:

Motivation: The motivation for this project stems from the need for Medicare patients, who are large users of medical services in the United States, to easily compare healthcare facilities when deciding where to receive services. Medicare offers a vast amount of quality data for all participating facilities, but patients cannot easily search for recommended facilities for specific concerns.

Project Goal: The goal of the project is for Medicare patients to quickly and easily access information and recommendations about which healthcare facilities provide high-quality care for common procedures. The web application will provide a comprehensive user-friendly destination for patients to quickly compare facilities and make better decisions about where to receive their care.

Proposed methods: This project will utilize publicly available data sets describing hospital performance reported in the state of New York from Medicare.gov and SPARCS. Amazon Web Services will host the database and MySQL will be used to manage the database. R's Shiny package will be used for the web application and Shiny server is used to host these web applications online.

Results: The outcome was a web application that Medicare patients can use to receive recommendations about which healthcare facilities will provide the highest quality care for their anticipated procedure. These recommendations are based upon criteria such as location, infection rates, death and complication rates, and readmission rates.

I. Background:

Ability to assess quality of care at hospitals:

Although there are currently many publicly available datasets presenting hospital quality information to consumers, they are typically not presented in a way that is meaningful. A 2015 study demonstrated that current methods of presenting hospital-acquired infection data, for example, were incorrectly interpreted by more than 40% of study participants [6]. Hospital rating services in general are more likely to confuse patients rather than inform them. This is due to their current lack of accessibility, consistency, and meaning [6]. Providing too much information to the consumer has also been shown to obscure otherwise meaningful comparisons and confuse consumers [11].

However, the quality measures used in ranking services must extend beyond readmission rates. This metric alone is not enough to accurately measure hospital quality [10]. Current quality rankings heavily focus on this metric because it is a major area of focus in recent policy, including value-based payment [10].

The measures that are used in these rating services use different structures and definitions, making them inconsistent and sometimes conflicting. A study performed by Rothberg and colleagues (2008) revealed the lack of agreement between various hospital ranking websites, even when using the same metric [11]. Proper synthesis and prioritization of quality measures is needed to make the hospital data accurate and meaningful for patients.

Existing analytics to assess hospital performances:

Most data collected by hospitals over patients who are admitted to hospitals are used by hospital management, physicians, and staff to improve on the care that is given at the facilities. Alternatively, looking to facilitate patients, US News and World Report, The Centers for Medicare and Medicaid Services, and The Leapfrog Group have gathered health data reported by hospitals and healthcare providers across the United States. These three entities have made public the analysis of these data via websites that can be accessed by patients looking for hospitals that have performed well in various areas of care and safety [3,7,9].

For all three entities previously discussed, the Centers for Medicare and Medicaid Services (CMS) data collection on more than 4,000 hospitals in the United States that are Medicare-certified, including Veterans Administration hospitals, have been utilized in the analysis of hospitals' performances [14]. Hospital Compare, funded and supported by CMS, use their data to inform Medicare patients about hospitals' performance concerning care as well as general information about services offered. The performance of each hospital is arranged into seven categories based on various metrics from readmittance rates after a procedure to results of patients' experiences at the facility [7]. However, these data that drive the hospital comparisons for the Medicare community are not based on a wide range of specialties to address various health conditions prevalent in the senior community.

Alternatively, US News and World Report has been publishing national and state hospital rankings that pertain to healthcare specialties described in one of 16 categories. These comparisons are data driven in 12 categories, “cancer, cardiology & heart surgery, diabetes & endocrinology, ear, nose & throat, gastroenterology & GI surgery, geriatrics, gynecology, nephrology, neurology & neurosurgery, orthopedics, pulmonology & lung surgery, and urology” [9]. Healthcare providers gave assessments that were used to rank facilities in the remaining 4 categories: ophthalmology, psychiatry, rehabilitation, and rheumatology [9]. Notably, these rankings can be focused into a large subset, usually at the state level if outside one of 89 large metropolitan areas, forcing potential patients to examine several facilities until those pertaining to a specific region are found [4].

Although Best Hospitals and Hospital Compare websites have filled the need for patients to have access to hospitals’ care performance, the actual analysis and reporting of the data has had issues. First most, these systems were found to have diverging scores for the many of the same hospitals [5]. Because both tools use different data sources as well as approaches to the data analysis, the disagreement over many hospitals’ performances is understandable [5]. Compounding the frustration with users, another study by Sinaiko et al demonstrated that but the inability to personalize or focus the metrics to future patients’ specifications in an understandable format was not attainable with the existing reports [12]. A further inspection of the data and analysis these reporting methods took revealed a change in hospitals performance over time in another study [10]. Measures that used readmission rates showed those hospitals in the lower quartiles for readmission rates migrated to higher quartiles and vice versa over a period of three years [10]. Therefore, qualifying a hospital’s performance by readmission rates following a procedure may need to be averaged over a certain number of years to gain a broader understanding of the metric.

Focusing on mortality and safety measures, The Leapfrog Group took a different approach with the publicly available Hospital Compare data sets. These sets along with data from The Joint Commission, The Leapfrog Group, and The Commonwealth Fund were used to calculate safety scores, representing safety measures and issues in a hospital setting for 2652 general acute care hospitals nationwide [3]. The data was transformed into relative safety scores weighted for strength of evidence (for safety issue occurrences), opportunity for improvement, and impact. The impact score was a combination of scores based on the number of patients affected by the measure and the “severity of harm” for individual patients. Each measure’s score was normalized by the hospital’s Z-score for that measure [3]. The limitations to the analysis given by this online tool include the small amount of publicly available data on the safety measures of each hospital, the error inherent in the data reported by hospital administration survey data (not clinical data), and the overestimate or understatement of a hospital’s performance from those with excluded measures [3].

The gaps in patient access to hospital performance:

Our project serves the patient population over the age of 60, which is a population estimated to grow 30% by 2025 (NAHU, 2015). There has been a need for databases that would cater to the Medicare population to make informed healthcare decisions. This project recommends hospitals based on their performance of commonly opted procedures in the state of New York via the SPARCS datasets. The SPARCS datasets accumulate measures based on a variety of medical procedures instead of one or two [13].

Despite numerous information systems available to the Medicare population, there is a lack of engagement with the systems due to the use of healthcare jargon [12]. To increase engagement, we hope to use nontechnical language and use clear statistical indexes to reduce the performance score to a single metric [11]. It has become increasingly clear over the years, that use of visualization techniques helps convey messages in a better manner.

One of the major issues with database accessibility via web lies with the interface compatibility on mobile devices. This project will accommodate the web interface to a mobile friendly architecture which will increase engagement with consumers. As the data used by the project is from the SPARCS datasets as compared to CMS data, a higher quality of data will be realized, which often is an issue when data is collected and made from various databases with differing metrics.

Furthermore, as there is growth in the aging population, this application could be expanded to keep record of the debilitating diseases like Alzheimer's and various subtypes of cancers across patients, which would assist in making clinical research decisions [8].

Contributions of the web-based hospital analysis:

This database-based web application will help patients to easily choose a hospital to receive care based upon comprehensive and accurate quality measures. Recommendations will be visually appealing and easy to understand for consumers. Technical information that is extraneous to the consumer’s needs will not be displayed to keep the recommendations as simple and meaningful as possible. Interpretation that was previously required from patients in traditional hospital comparison websites will no longer be needed. This data will not only be useful to help patients find high-quality care, but it can also be used by hospitals themselves to easily understand how their facility is ranked among other hospitals. This can be used to drive quality performance improvement initiatives.

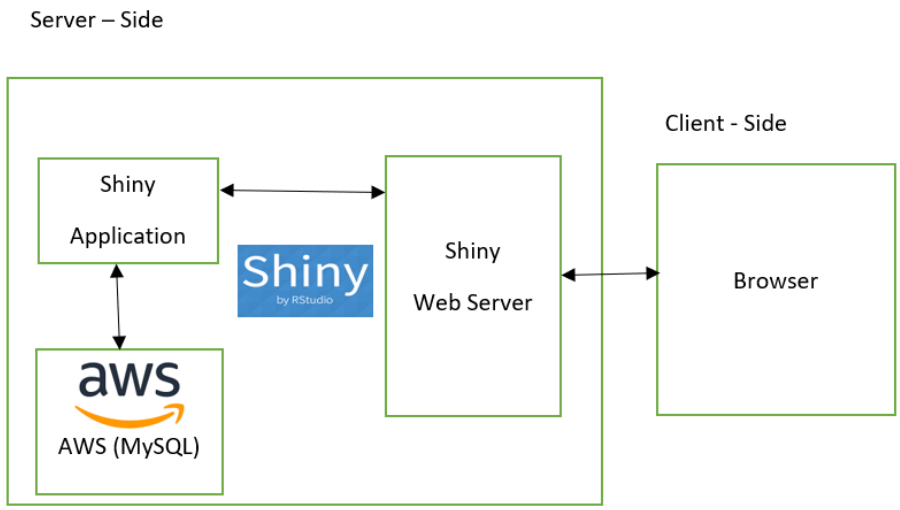
II. Methods:

Amazon web services, specifically the amazon relation database, is free for a year which can be used in combination with MySQL in order to host a database. AWS was chosen due to its speed, scalability, and security. Hosting a database on a cloud makes it easier to administer and centrally upgrade it.

R-Shiny was chosen as the tool to develop the web application. Shiny is a simple framework that allows for web application development using R without requiring additional web development toolsets. It was chosen because of its ease-of-use and ability to create interactive data visualizations. The leaflet package that can be used to display geographical data is also functional

Apart from being able to use R shiny to make web application, R shiny.io server also lets us host Shiny applications over the web. One of the main advantages with using open source tool like R shiny is its ability to be user friendly which helps users with no prior knowledge of firewall, yet it can access AWS securely to read and write data.

Diagram 1: the two-tier architecture of this application



Data

The healthcare data used was the result of a cooperation between New York healthcare providers and the state of New York. This cooperative, SPARCS, provides datasets

reporting hospital's performance for a variety of measures. The datasets that we implemented aggregated the data into three categories from measures gathered from 2009-2017: inpatient quality measures, patient safety measures, and prevention quality measures. The rates for each measure were aggregated to return three top performing hospitals for each category which. Subsequently, data scraped for New York state hospitals' locations (latitude, longitude) from www.lat-long.com is able to provide a map using Leaflet © to display the location for the top performing hospitals in each category.

The inpatient quality measures were described by the number of "ambulatory care sensitive conditions" treated at each facility [1]. The conditions treated are those for which either long-term or short-term care can either prevent, to some extent, disease progression or complications. To give more significance to morality rates that are directly related to surgical outcomes, these rates will be multiplied by 2. The measures related to mortality rates involving illness or injury and typically misused surgical procedures will be multiplied by 1.

Table 1: The IQI values are described with their corresponding weights.

IQI_Code	IQI_Value	weight
IQI08	Esophageal Resection Mortality Rate	2
IQI09	Pancreatic Resection Mortality Rate	2
IQI09A	Pancreatic Resection Mortality Rate Stratum A: Presence of Pancreatic Cancer	2
IQI09B	Pancreatic Resection Mortality Rate Stratum B: Absence of Pancreatic Cancer	2
IQI11	Abdominal Aortic Aneurysm (AAA) Repair Mortality Rate	2
IQI11A	Abdominal Aortic Aneurysm (AAA) Repair Mortality Rate - Ruptured Open	2
IQI11B	Abdominal Aortic Aneurysm (AAA) Repair Mortality Rate - Unruptured Open	2
IQI11C	Abdominal Aortic Aneurysm (AAA) Repair Mortality Rate - Ruptured Endovascular	2
IQI11D	Abdominal Aortic Aneurysm (AAA) Repair Mortality Rate - Unruptured Endovascular	2
IQI13	Craniotomy Mortality Rate	2
IQI14	Hip Replacement Mortality Rate	2
IQI15	Acute Myocardial Infarction (AMI) Mortality Rate	2
IQI16	Heart Failure Mortality Rate	2
IQI18	Gastrointestinal Hemorrhage Mortality Rate	2
IQI19	Hip Fracture Mortality Rate	2
IQI20	Pneumonia Mortality Rate	2
IQI23	Laparoscopic Cholecystectomy Rate	1
IQI24	Incidental Appendectomy in the Elderly Rate	1
IQI25	Bilateral Cardiac Catheterization Rate	1
IQI31	Carotid Endarterectomy Mortality Rate	2
IQI32	Acute Myocardial Infarction (AMI) Mortality Rate, Without Transfer Cases	2

The second category of data reports safety measures on infection, injury, and death that have resulted from the treatment received at each facility. These rates will give an understanding of the hospital's efficacy in safety protocols for patient care. The severity of the safety measures that each hospital reported varied. Therefore, to give more

significance to the safety rates that have an impact on patients' survival, each measure was weighted as was done with the quality indicators. These weight values varied between 1 and 2.

Table 2: The PSI values with their corresponding weights

PSI Code	PSI Value	weight
PSI02	Death Rate in Low-Mortality Diagnosis Related Groups (DRGs)	2
PSI03	Pressure Ulcer Rate	1.1
PSI04	Death Rate among Surgical Inpatients with Serious Treatable Conditions	2
PSI04A	(DRASISTC) Stratum 04A: Deep Vein Thrombosis/Pulmonary Embolism (DVT/PE)	2
PSI04B	(DRASISTC) Stratum 04B: Pneumonia	2
PSI04C	(DRASISTC) Stratum 04C: Sepsis	2
PSI04D	(DRASISTC) Stratum 04D: Shock/Cardiac Arrest	2
PSI04E	(DRASISTC) Stratum 04E: Gastrointestinal (GI) Hemorrhage/Acute Ulcer	2
PSI06	Iatrogenic Pneumothorax Rate	1.7
PSI07	Central Venous Catheter-Related Blood Stream Infection Rate	1.3
PSI08	In Hospital Fall with Hip Fracture Rate	1.5
PSI09	Perioperative Hemorrhage or Hematoma Rate	1.6
PSI10	Postoperative Acute Kidney Injury Requiring Dialysis Rate	1.6
PSI11	Postoperative Respiratory Failure Rate	1.8
PSI12	Perioperative Pulmonary Embolism or Deep Vein Thrombosis Rate	1.2
PSI13	Postoperative Sepsis Rate	1.2
PSI14	Postoperative Wound Dehiscence Rate	1.2
PSI14A	Postoperative Wound Dehiscence Rate Stratum 14A: Open Approach	1.2
PSI14B	Postoperative Wound Dehiscence Rate Stratum 14B: Non-Open Approach	1.2
PSI15	Unrecognized Abdominopelvic Accidental Puncture or Laceration Rate	1.4

Lastly, the prevention quality measures will be defined with rates reported by facilities for potentially preventable complications and potentially preventable readmissions. The potentially preventable readmissions rates were reported for patients that were discharged from a facility within 30 days of readmission. These readmissions are attributed to inadequate follow-up care after the initial discharge or inadequate care prior to discharge. The possible preventable complication rates are obtained from negative outcome counts resulting in care or treatment given by the reporting facility during hospitalization as opposed to the progression of a disorder or disease. These possible preventable rates were averaged to be used to rank facilities based on their abilities to prevent negative outcomes for patients.

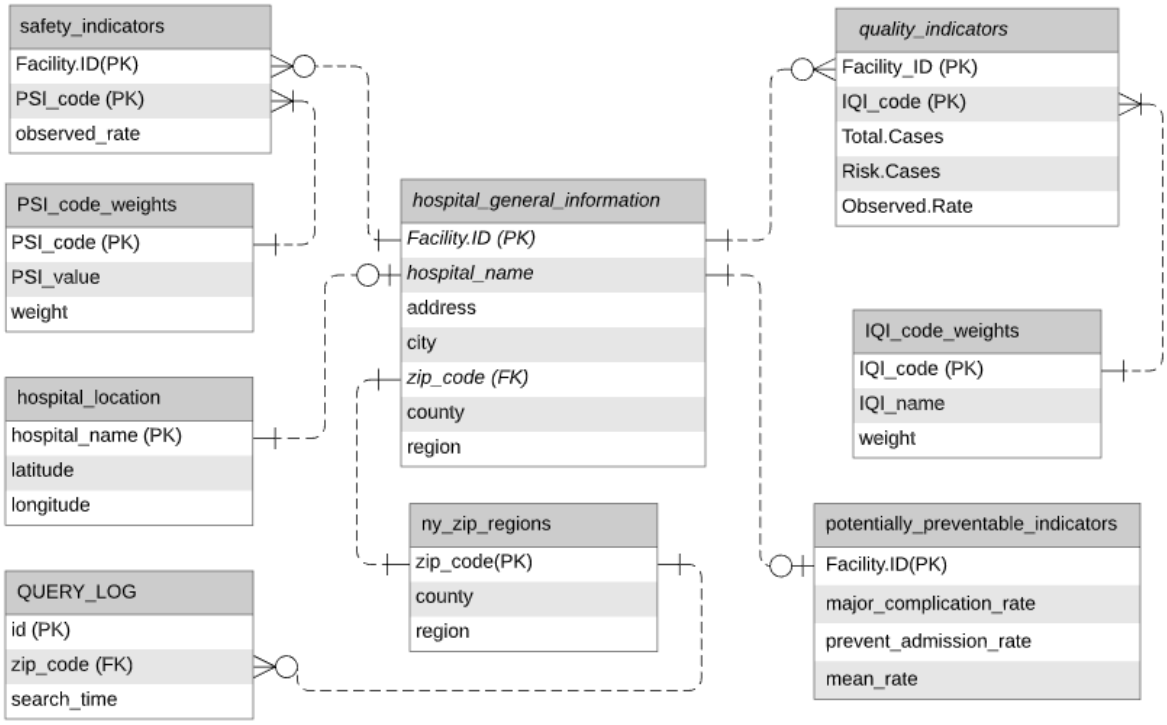
Because the hospitals reporting data for each area code did not include more than three in many areas, New York was divided into 14 areas with a list of zip codes pertaining to that area. A comprehensive list of zip codes for each area was obtained from data.ny.gov to create a table to query in the database. The hospitals in these areas

were compared to give a recommendation for a zip code search. Although the area a patient may have to travel for care increased, better comparison for recommendations was achieved by this aggregation.

Hyperlinks were also incorporated into the markers for the map displaying the location of the recommended hospitals. These links would give patients a google search by which more information can be acquired in the search for quality care.

Lastly, to assess the usage of the apps, a query table was created that the app writes time stamps for each zip code searched. The table will allow time series analysis for the apps usage to determine when and were there is a demand by patients wanting quality care.

Diagram 2: ERD for Hospital Recommendations



III. Results:

The results of a zip code query pulled at most three hospitals to recommend for each parameter. These searches can result in some overlap of hospital recommendations if a high performing hospital is in the queried area. A map with three markers allows patients to see roads and landmarks around recommended hospitals along with the names and addresses displayed in a table above the map. The patients can choose which parameter to examine by a tab with the parameter label. The marker for each hospital on the map in the tabbed page opens to a search for further investigation into the chosen hospital.

The usage tab on the page allows users and developers to examine the areas of the most demand for the application. Future research may reveal the time and place for the most demand by Medicare patients for quality care.

Figure 1: Quality results for a zip code query

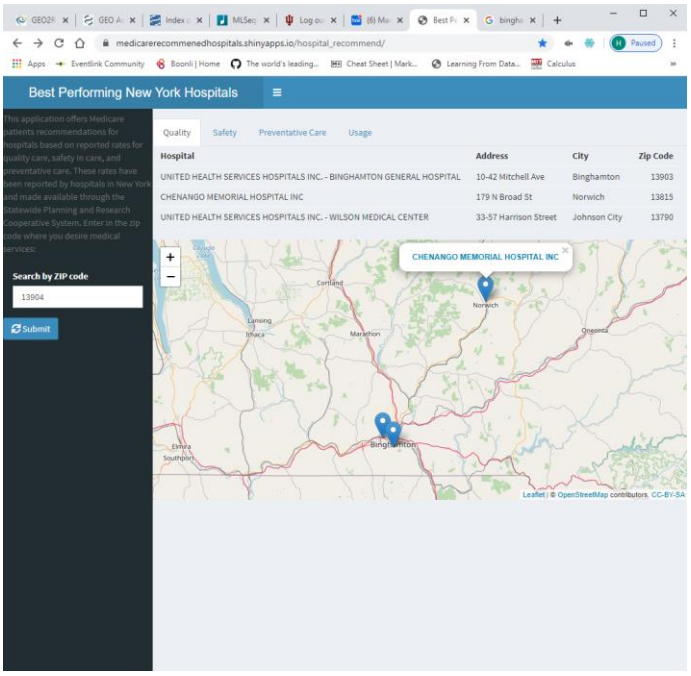


Figure 2: Safety results for a zip code query

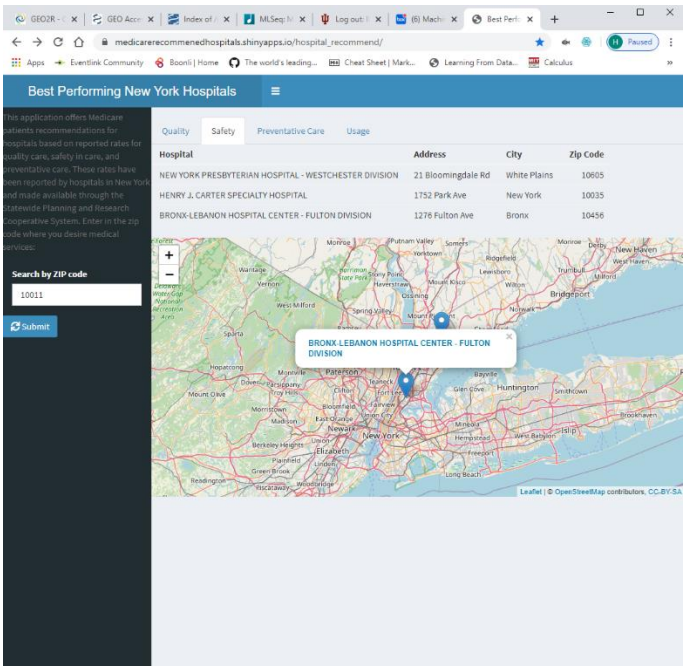


Figure 3: Preventative Care results for a zip code query

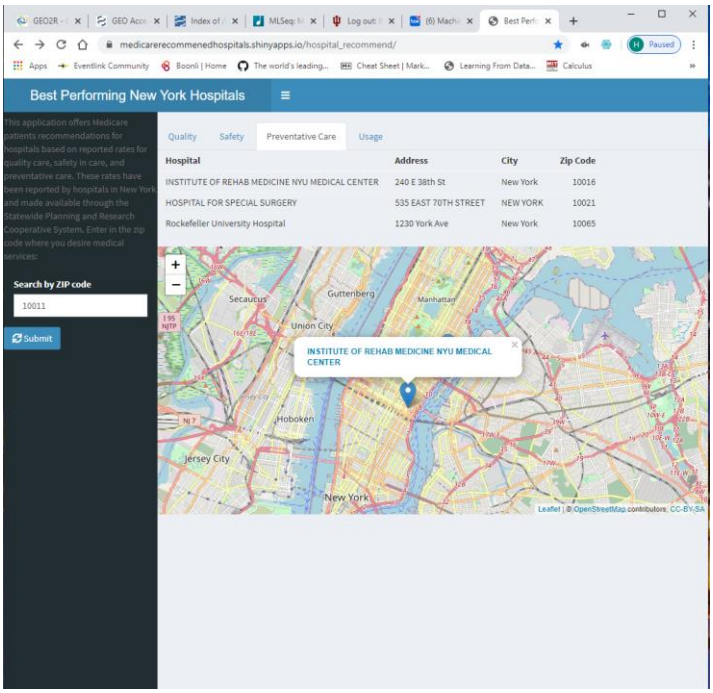


Figure 4: A Hyperlinked Google search for a chosen hospital

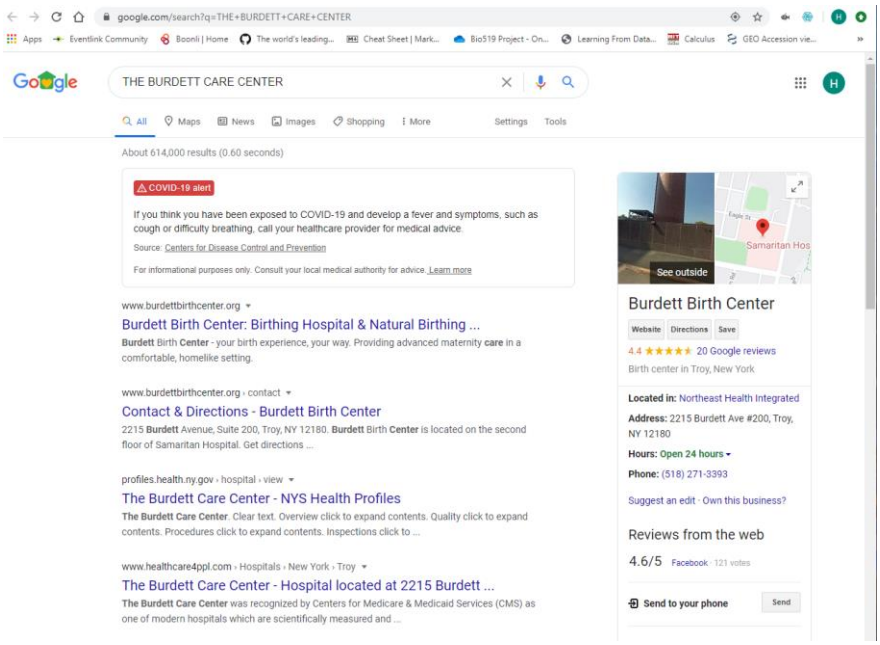
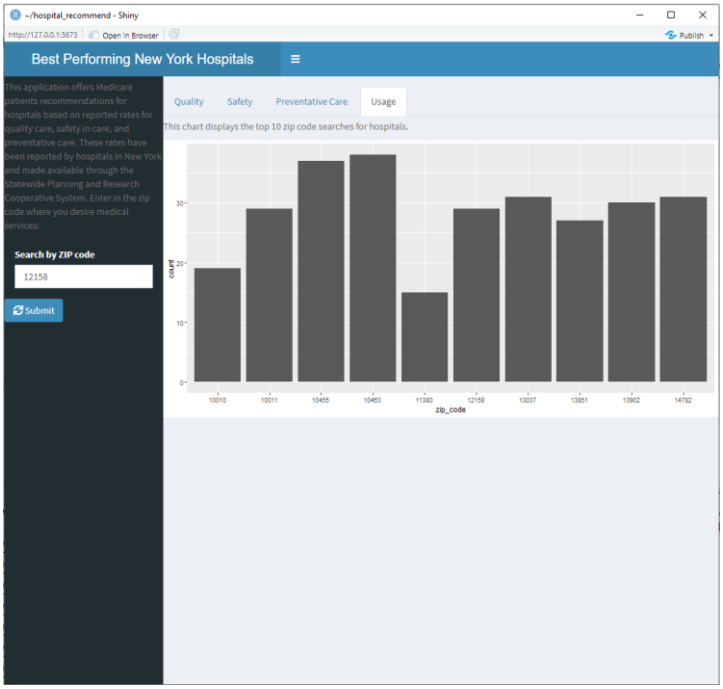


Figure 5: A bar graph of the most queried zip codes for the application



Discussion:

While the original scope of the project was intended to cover hospitals across the United States, our group decided to narrow the scope to include New York hospitals only. This decision was based upon the lack of consistent hospital data across all states. It was also more feasible to test our project with only one state's data and then expand to a greater geographic area in future studies. The data contained in the SPARCS database has a smaller scope and is simpler to compile for use in the database.

Although the SPARCS data was simpler to compile, there were still challenges regarding cleaning the data files and determining how to organize the ER diagram. A determination had to be made about how to compare hospitals and weights had to be assigned to the safety and quality indicators to properly rank the hospitals. This required studying the indicators and determining their relative influence on patient mortality. Though one overall ranking may be ideal, this is difficult in practice because each patient will have different priorities. Therefore, we decided to display the top 3 hospitals in each of the three areas: inpatient quality, safety, and prevention quality.

An additional challenge in implementing this application was the R Shiny packages' limitations. The functionality of built in widgets and functions did not allow for a high level of customization for the user interface. To overcome this issue, a broader knowledge of HTML, CSS, and java script would allow augmentation to the preexisting functionalities of the packages.

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